

Monge Formulation

Statements, hints and worked solutions

Each exercise is followed by question-by-question hints and a worked solution. Try the questions before reading the hints.

Exercise 1 (Discrete assignment).

Let $n = 3$ and $C = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 2 \\ 4 & 3 & 1 \end{pmatrix}$. Identify every optimal assignment.

Useful hints

Enumerate the six permutations of $\{1, 2, 3\}$ and sum one matrix entry per row and column.

Detailed solution

For a permutation σ , the cost is $C(1, \sigma(1)) + C(2, \sigma(2)) + C(3, \sigma(3))$. The minimum is 5, achieved by the two assignments

$$(1 \rightarrow 1, 2 \rightarrow 2, 3 \rightarrow 3) \quad \text{and} \quad (1 \rightarrow 2, 2 \rightarrow 1, 3 \rightarrow 3).$$

All others have cost 7 or 9.

Exercise 2 (Two points to two points: when does each permutation win?).

Fix $x_1, y_1, y_2 \in \mathbb{R}^d$ with $y_1 \neq y_2$ and let $x_2 = z \in \mathbb{R}^d$ vary. With quadratic cost $c(x, y) = \|x - y\|^2$, compare the two assignments

$$\text{Id: } x_1 \mapsto y_1, z \mapsto y_2 \quad \text{vs} \quad \text{Swap: } x_1 \mapsto y_2, z \mapsto y_1.$$

Show that the set of z where Id is no more expensive than Swap is a half-space bounded by a hyperplane orthogonal to $y_1 - y_2$. Give the equation of this hyperplane and explain that it divides $\mathbb{R}^d$ into two regions, one for each optimal permutation.

Useful hints

Expand the difference of the two costs; the terms $\|z\|^2$ and the target squared norms cancel.

Detailed solution

The two total costs differ by

$$\text{Id}(z) - \text{Swap}(z) = 2\langle z - x_1, y_1 - y_2 \rangle.$$

Thus Id is optimal when $\langle z - x_1, y_1 - y_2 \rangle \leq 0$, Swap is optimal when this quantity is nonnegative, and both tie on the hyperplane $\{z : \langle z - x_1, y_1 - y_2 \rangle = 0\}$. Strict inequalities give strict preference. The assumption $y_1 \neq y_2$ ensures this is a genuine hyperplane; if $y_1 = y_2$, both assignments always tie.

Exercise 3 (Discrete $\rightarrow$ continuous and continuous $\rightarrow$ discrete).

- (1) In 1D let $\alpha = \sum_{i=1}^N a_i \delta_{x_i}$ (finite support) and let β be absolutely continuous. Show that there is no measurable map T with $T_{\#}\alpha = \beta$.
- (2) Conversely, let α be absolutely continuous on $\mathbb{R}$ and let $\beta = \sum_{j=1}^M b_j \delta_{y_j}$ with $\sum_j b_j = 1$. Build maps T (many, up to α -a.e. equality, if β has at least two distinct atoms of positive mass) with $T_{\#}\alpha = \beta$.
- (3) Let $\alpha = \mathcal{U}[0, 1]$ and let $\beta = \sum_{j=1}^M b_j \delta_{y_j}$ on $\mathbb{R}$ with $b_j > 0$, $\sum_j b_j = 1$, and $y_1 < \dots < y_M$. For $p \geq 1$, compute an $\mathcal{W}_p$ -optimal Monge map $T : [0, 1] \rightarrow \mathbb{R}$ from α to β .
- (4) Suppose $b_j = 1/n$ for $j = 1, \dots, n$ and $y_j = \frac{j-1/2}{n}$ (midpoints of the n equal subintervals of $[0, 1]$). Compute $W_2^2(\alpha, \beta)$.

Useful hints

- (1) A pushforward cannot split the mass of an atom.
- (2) Partition the source into measurable sets with masses b_j ; quantiles provide one such partition.
- (3) The interval sent to y_j must have length b_j . Use increasing rearrangement; uniqueness need not hold when $p = 1$.
- (4) Compute the squared distance to the midpoint in one cell, then multiply by n .

Detailed solution

- (1) *No map from discrete to continuous.* If $\alpha = \sum_{i=1}^N a_i \delta_{x_i}$ and T is measurable, then $T_{\#}\alpha = \sum_{i=1}^N a_i \delta_{T(x_i)}$, which is purely atomic; it cannot equal an absolutely continuous β .
- (2) *Many maps from continuous to discrete.* Let α be absolutely continuous and $\beta = \sum_{j=1}^M b_j \delta_{y_j}$. Choose any measurable partition $\{A_j\}_{j=1}^M$ with $\alpha(A_j) = b_j$ (e.g. cut by quantiles) and set $T = \sum_{j=1}^M y_j \mathbf{1}_{A_j}$. Then $T_{\#}\alpha = \beta$. Partitions differing on a set of positive α -mass give distinct maps when the target has at least two distinct positive-mass atoms. A one-atom target forces the constant map.
- (3) *Optimal map from $\mathcal{U}[0, 1]$ to a discrete law.* Let $F_{\alpha}(x) = x$ on $[0, 1]$ and F_{β} be the CDF of β , with partial sums $s_j = \sum_{k=1}^j b_k$ and $s_0 = 0$. The quantile of β is $F_{\beta}^{-1}(u) = y_j$ for $u \in (s_{j-1}, s_j]$. In 1D an $\mathcal{W}_p$ -optimal Monge map is the increasing rearrangement:

$$T = F_{\beta}^{-1} \circ F_{\alpha} = F_{\beta}^{-1}, \quad T(x) = y_j \text{ for } x \in (s_{j-1}, s_j].$$

- (4) *Uniform atoms at midpoints:* $b_j = \frac{1}{n}$, $y_j = \frac{j-1/2}{n}$. Here $s_j = \frac{j}{n}$, so by (3) the optimal map is $T(x) = \frac{j-1/2}{n}$ for $x \in (\frac{j-1}{n}, \frac{j}{n}]$. Then $W_2^2(\alpha, \beta) = \sum_{j=1}^n \int_{(j-1)/n}^{j/n} \left(x - \frac{j-1/2}{n}\right)^2 dx$. Shift to the midpoint of each cell: with $u = x - \frac{j-1/2}{n}$ we integrate over $u \in (-\frac{1}{2n}, \frac{1}{2n}]$:

$$\int_{(j-1)/n}^{j/n} \left(x - \frac{j-1/2}{n}\right)^2 dx = \int_{-1/(2n)}^{1/(2n)} u^2 du = \left[\frac{u^3}{3}\right]_{-1/(2n)}^{1/(2n)} = \frac{2}{3} \left(\frac{1}{2n}\right)^3 = \frac{1}{12n^3}.$$

Summing over $j = 1, \dots, n$ gives $W_2^2(\alpha, \beta) = n \cdot \frac{1}{12n^3} = \frac{1}{12n^2}$. Equivalently, by the 1D quantile formula $W_2^2(\alpha, \beta) = \sum_{j=1}^n \int_{(j-1)/n}^{j/n} \left|t - \frac{j-1/2}{n}\right|^2 dt = \frac{1}{12n^2}$.

Exercise 4 (Uniform laws on intervals, quadratic cost).

Let $\alpha = \mathcal{U}_{[a,b]}$ and $\beta = \mathcal{U}_{[c,d]}$ on $\mathbb{R}$ with $a < b, c < d$, and $c(x, y) = |x - y|^2$.

- (i) Find the Monge optimal transport map T .
- (ii) Compute $\mathcal{W}_2^2(\alpha, \beta)$ in closed form.

Useful hints

- (i) Match equal cumulative probabilities.
- (ii) Express each quantile as its mean plus its interval length times $t - 1/2$; the mixed term integrates to zero.

Detailed solution

(i) In 1D the optimal map is the increasing rearrangement: $T(x) = c + \frac{d-c}{b-a}(x - a)$. (ii) Using quantiles,

$$\mathcal{W}_2^2 = (m_\alpha - m_\beta)^2 + \frac{(\ell_\alpha - \ell_\beta)^2}{12}, \quad m_\alpha = \frac{a+b}{2}, \quad m_\beta = \frac{c+d}{2}, \quad \ell_\alpha = b - a, \quad \ell_\beta = d - c.$$

Indeed, the quantile difference is $(m_\alpha - m_\beta) + (\ell_\alpha - \ell_\beta)(t - 1/2)$. Its square integrates to the displayed expression because $\int_0^1 (t - 1/2) dt = 0$ and $\int_0^1 (t - 1/2)^2 dt = 1/12$.

Exercise 5 (Many Monge solutions for $|x - y|$).

Let $\alpha = \mathcal{U}_{[0,2]}$, $\beta = \mathcal{U}_{[1,3]}$ and $c(x, y) = |x - y|$.

- (i) Show $T_1(x) = x + 1$ is optimal and $\mathcal{W}_1(\alpha, \beta) = 1$.
- (ii) Show

$$T_2(x) = \begin{cases} x + 2, & x \in [0, 1], \\ x, & x \in (1, 2], \end{cases}$$

is also optimal.

- (iii) Build a family of additional Monge solutions.

Useful hints

- (i) Any plan has cost at least the difference of the means.
- (ii) Check the images of the two half-intervals separately.
- (iii) Equality in $|y - x| \geq y - x$ holds when $y \geq x$. Construct many measure-preserving maps with this property.

Detailed solution

(i) Every coupling satisfies $\int |y - x| d\pi \geq \int (y - x) d\pi = 2 - 1 = 1$. The translation $T_1(x) = x + 1$ has the required image and attains this bound.

(ii) The first half of $[0, 2]$ is translated onto $[2, 3]$, and the second half is left on $[1, 2]$. Both carry density $1/2$, so T_2 has image β . Since $T_2(x) \geq x$, its cost is again 1.

(iii) Let $A \subset [0, 2]$ be measurable with Lebesgue measure 1, and define $F_A(x) = |A \cap [0, x]|$ and $F_{A^c}(x) = x - F_A(x)$. Set

$$T_A(x) = \begin{cases} 1 + F_A(x), & x \in A, \\ 2 + F_{A^c}(x), & x \in [0, 2] \setminus A. \end{cases}$$

The cumulative transform of the atomless probability $\mathbf{1}_A dx$ is uniform, so these two restrictions push forward to Lebesgue measure on $[1, 2]$ and $[2, 3]$. Hence $(T_A)_\# \alpha = \beta$. On A , $T_A(x) - x = 1 - F_{A^c}(x) \geq 0$; on A^c , $T_A(x) \geq 2 \geq x$. Thus all these maps attain cost 1. There are also non-deterministic optimal couplings: for $0 < \theta < 1$, mix the graph plans of T_1 and T_2 with weights θ and $1 - \theta$.

Exercise 6 (Affine images).

(1) **1-D case.** Let $\mu \in \mathcal{P}_2(\mathbb{R})$ and $\nu = (x \mapsto ax + b)_\# \mu$ with $a > 0$. What is the Monge map from μ to ν for the quadratic cost? Compute the W_2 distance in terms of the mean $m_\mu = \int x d\mu$ and variance $\text{Var}(\mu) = \int (x - m_\mu)^2 d\mu$.

(2) **Higher dimension.** Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and $\nu = (x \mapsto Ax + b)_\# \mu$ with $A \in \mathbb{R}^{d \times d}$. Give a simple condition on A ensuring that $T(x) = Ax + b$ is the Brenier map (assume μ is absolutely continuous). Under this condition, compute the W_2 distance in terms of $m_\mu = \mathbb{E}[X]$ and the covariance $\Sigma_\mu = \mathbb{E}[(X - m_\mu)(X - m_\mu)^\top]$.

Useful hints

- (1) A positive-slope affine map is increasing. Expand $|(1 - a)X - b|^2$ around the mean.
- (2) A symmetric positive-semidefinite matrix gives a convex quadratic potential. Use $\mathbb{E}\|MX\|^2 = \text{tr}(M\Sigma_\mu M^\top) + \|Mm_\mu\|^2$.

Detailed solution

(1) **1D.** Since $a > 0$, $x \mapsto ax + b$ is increasing. In 1D the $\mathcal{W}_2$ -optimal Monge map is the monotone rearrangement, hence $T(x) = ax + b$. Let $X \sim \mu$ with mean m_μ and variance $\text{Var}(\mu)$. Then

$$\mathcal{W}_2^2(\mu, \nu) = \mathbb{E}[(X - (aX + b))^2] = \mathbb{E}[(1 - a)X - b]^2.$$

Expand around the mean:

$$(1 - a)X - b = (1 - a)(X - m_\mu) + ((1 - a)m_\mu - b),$$

so the cross term vanishes and

$$\mathcal{W}_2^2(\mu, \nu) = (1 - a)^2 \text{Var}(\mu) + ((1 - a)m_\mu - b)^2.$$

(2) $\mathbb{R}^d$. If A is **symmetric positive semidefinite**, then there exists a convex quadratic potential $\phi(x) = \frac{1}{2}x^\top Ax + b \cdot x + c$ with $\nabla \phi(x) = Ax + b$. By Brenier, for absolutely continuous μ , $T(x) = Ax + b$ is the unique (a.e.) $\mathcal{W}_2$ -optimal map.

Let $X \sim \mu$, mean m_μ , covariance Σ_μ . Write

$$X - (AX + b) = (I - A)(X - m_\mu) + ((I - A)m_\mu - b).$$

Taking squared norm and expectation, the cross term drops and

$$\mathbb{E}\|X - (AX + b)\|^2 = \text{tr}((I - A)\Sigma_\mu(I - A)^\top) + \|(I - A)m_\mu - b\|^2.$$

Hence

$$\mathcal{W}_2^2(\mu, \nu) = \text{tr}((I - A)\Sigma_\mu(I - A)^\top) + \|(I - A)m_\mu - b\|^2 \quad (\text{for } A \geq 0 \text{ symmetric}).$$

Exercise 7 (Piecewise constant densities on $[0,1]$).

, quadratic cost] Let α have density $f = 1$ on $[0, \frac{1}{2})$ and $f = 2$ on $[\frac{1}{2}, 1]$, while β has density $g = 2$ on $[0, \frac{1}{2})$ and $g = 1$ on $[\frac{1}{2}, 1]$. Construct (and sketch) the optimal Monge map on $\mathbb{R}$ for the quadratic cost and compute the exact cost (for the corresponding probability measures obtained by normalising by the common mass $3/2$).

Useful hints

Normalize both densities, compute their cumulative functions, and find where the source cumulative crosses the target breakpoint $2/3$. Integrate the squared displacement on the resulting three intervals.

Detailed solution

After normalisation, the densities are $\hat{f} = \frac{2}{3}$ on $[0, \frac{1}{2})$, $\hat{f} = \frac{4}{3}$ on $[\frac{1}{2}, 1]$, and $\hat{g} = \frac{4}{3}$ on $[0, \frac{1}{2})$, $\hat{g} = \frac{2}{3}$ on $[\frac{1}{2}, 1]$. The optimal map is $T = F_{\hat{\beta}}^{-1} \circ F_{\hat{\alpha}}$:

$$T(x) = \begin{cases} \frac{x}{2}, & x \in [0, \frac{1}{2}], \\ x - \frac{1}{4}, & x \in [\frac{1}{2}, \frac{3}{4}], \\ 2x - 1, & x \in [\frac{3}{4}, 1]. \end{cases}$$

The graph consists of three continuous line segments of slopes $1/2, 1, 2$, joining $(0, 0)$, $(1/2, 1/4)$, $(3/4, 1/2)$, and $(1, 1)$. The cost for the normalized measures is

$$\frac{2}{3} \int_0^{1/2} \frac{x^2}{4} dx + \frac{4}{3} \int_{1/2}^{3/4} \frac{1}{16} dx + \frac{4}{3} \int_{3/4}^1 (1-x)^2 dx = \frac{1}{144} + \frac{1}{48} + \frac{1}{144} = \frac{5}{144}.$$

Exercise 8 (Translation decomposition for $\mathcal{W}_2$).

Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ (finite second moments). For $u \in \mathbb{R}^d$ write $T_u(x) = x + u$.

- (a) Show there exist unique vectors a, b such that the *centred* measures

$$\alpha_0 := (T_{-a})_{\#}\alpha, \quad \beta_0 := (T_{-b})_{\#}\beta$$

satisfy $\int x d\alpha_0(x) = \int x d\beta_0(x) = 0$, and

$$\alpha = (T_a)_{\#}\alpha_0, \quad \beta = (T_b)_{\#}\beta_0.$$

- (b) Prove the decomposition

$$\mathcal{W}_2^2(\alpha, \beta) = \|a - b\|^2 + \mathcal{W}_2^2(\alpha_0, \beta_0).$$

- (c) If α is absolutely continuous and T_0 is the Brenier map from α_0 to β_0 , show that

$$T(x) := T_0(x - a) + b$$

is the Brenier map from α to β .

Useful hints

- (a) Compute the mean of a translated measure.
- (b) Translate both coordinates of a coupling; the mixed term vanishes after integration, not pointwise.
- (c) Translate the source before applying T_0 , and translate its output afterwards.

Detailed solution

(a) *Centring and uniqueness.* Define $a = \int x d\alpha$ and $b = \int x d\beta$ (these exist by the $\mathcal{P}_2$ assumption). Put $\alpha_0 = (T_{-a})_{\#}\alpha$. Then, by change of variables,

$$\int x d\alpha_0(x) = \int (x - a) d\alpha(x) = \left(\int x d\alpha(x) \right) - a = 0.$$

Similarly $\int x d\beta_0(x) = 0$. Also $(T_a)_{\#}\alpha_0 = \alpha$ and $(T_b)_{\#}\beta_0 = \beta$ by construction. If $\tilde{a}$ also centres α , then $0 = \int x d(T_{-\tilde{a}})_{\#}\alpha = \int (x - \tilde{a}) d\alpha = \left(\int x d\alpha \right) - \tilde{a}$, hence $\tilde{a} = a$; likewise for b .

(b) *Decomposition of $\mathcal{W}_2^2$.* Let $\Pi(\mu, \nu)$ be the set of couplings with marginals μ, ν . Consider the bijection

$$\Phi : \Pi(\alpha_0, \beta_0) \longrightarrow \Pi(\alpha, \beta), \quad \Phi(\pi_0) := (T_a \times T_b)_{\#}\pi_0,$$

with inverse $\Phi^{-1}(\pi) = (T_{-a} \times T_{-b})_{\#}\pi$. For any $\pi_0 \in \Pi(\alpha_0, \beta_0)$,

$$\int \|x - y\|^2 d\Phi(\pi_0)(x, y) = \int \|(x + a) - (y + b)\|^2 d\pi_0(x, y).$$

Here is where the quadratic cost is used: expand the square,

$$\|(x + a) - (y + b)\|^2 = \|x - y\|^2 + \|a - b\|^2 + 2\langle a - b, x - y \rangle.$$

Integrating w.r.t. π_0 ,

$$\int \|(x + a) - (y + b)\|^2 d\pi_0 = \int \|x - y\|^2 d\pi_0 + \|a - b\|^2 + 2\left\langle a - b, \int (x - y) d\pi_0 \right\rangle.$$

Since π_0 has marginals α_0, β_0 ,

$$\int (x - y) d\pi_0 = \int x d\alpha_0 - \int y d\beta_0 = 0 - 0 = 0,$$

so the cross term vanishes and

$$\int \|(x + a) - (y + b)\|^2 d\pi_0 = \|a - b\|^2 + \int \|x - y\|^2 d\pi_0.$$

Taking the infimum over $\pi_0 \in \Pi(\alpha_0, \beta_0)$ yields

$$\mathcal{W}_2^2(\alpha, \beta) = \inf_{\pi \in \Pi(\alpha, \beta)} \int \|x - y\|^2 d\pi = \|a - b\|^2 + \inf_{\pi_0 \in \Pi(\alpha_0, \beta_0)} \int \|x - y\|^2 d\pi_0 = \|a - b\|^2 + \mathcal{W}_2^2(\alpha_0, \beta_0).$$

(c) *Relation between the optimal maps.* Assume α is absolutely continuous so the Brenier maps exist and are unique a.e. Let T_0 be the Brenier map $\alpha_0 \rightarrow \beta_0$. Define $T(x) = T_0(x - a) + b$. Then

$$T_{\#}\alpha = (T_0(\cdot - a) + b)_{\#}\alpha = (T_0 + b)_{\#}((T_{-a})_{\#}\alpha) = (T_0 + b)_{\#}\alpha_0 = (T_b)_{\#}(T_0)_{\#}\alpha_0 = (T_b)_{\#}\beta_0 = \beta.$$

Moreover, for $X \sim \alpha$ with $X - a \sim \alpha_0$,

$$\mathbb{E}\|X - T(X)\|^2 = \mathbb{E}\|(X - a) - T_0(X - a)\|^2 + \|a - b\|^2$$

(the cross term vanishes in expectation as above), so T attains the cost $\|a - b\|^2 + \mathcal{W}_2^2(\alpha_0, \beta_0)$. By part (b) this equals $\mathcal{W}_2^2(\alpha, \beta)$, hence T is optimal. Since T is the gradient of the convex potential $x \mapsto \phi_0(x - a) + b \cdot x$ (when $T_0 = \nabla \phi_0$), it is the Brenier map $\alpha \rightarrow \beta$.

Exercise 9 (Adding separable terms; quadratic cost vs. $-\langle x, y \rangle$).

Fix $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and consider Monge maps T with $T_{\#}\alpha = \beta$.

- (a) Let $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a measurable cost and define $\tilde{c}(x, y) = c(x, y) + u(x) + v(y)$, with $u : \mathbb{R}^d \rightarrow \mathbb{R}, v : \mathbb{R}^d \rightarrow \mathbb{R}$. Assume $u \in L^1(\alpha), v \in L^1(\beta)$, and that the integral of c is well-defined for every admissible map (for instance $c \geq 0$). Show that the sets of Monge minimizers for c and for $\tilde{c}$ (with the same marginals) coincide.
- (b) Let $\lambda > 0$ and define $\hat{c}(x, y) = \lambda c(x, y)$. Show that the Monge minimizers for c and $\hat{c}$ coincide.
- (c) Take $c_{\text{quad}}(x, y) = \|x - y\|^2$, $c_{\text{lin}}(x, y) = -\langle x, y \rangle$. Prove that c_{quad} and c_{lin} have the same Monge minimizers (for fixed α, β).

Useful hints

- (a) Use $\int v(T(x)) d\alpha = \int v d\beta$.
- (b) A positive multiplier preserves ordering.
- (c) Expand the square and apply the first two parts.

Detailed solution

(a) *Separable perturbations.* For any admissible map T with $T_{\#}\alpha = \beta$,

$$\int \tilde{c}(x, T(x)) d\alpha(x) = \int c(x, T(x)) d\alpha(x) + \underbrace{\int u(x) d\alpha(x)}_{\text{depends only on } \alpha} + \underbrace{\int v(y) d\beta(y)}_{\text{depends only on } \beta}.$$

The last two terms are independent of T , so the ordering of admissible maps by total cost is the same for c and $\tilde{c}$. Hence the minimizers coincide.

(b) *Positive rescaling.* For any admissible T ,

$$\int \hat{c}(x, T(x)) d\alpha(x) = \lambda \int c(x, T(x)) d\alpha(x).$$

Multiplying by $\lambda > 0$ preserves the argmin, so the sets of minimizers are the same.

(c) *Quadratic cost vs. $-\langle x, y \rangle$.* Decompose

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y \rangle = u(x) + v(y) + \tilde{c}(x, y),$$

with $u(x) = \|x\|^2$, $v(y) = \|y\|^2$, and $\tilde{c}(x, y) = -2\langle x, y \rangle$. By part (a), c_{quad} and $\tilde{c}$ have the same Monge minimizers. By part (b), $\tilde{c} = -2\langle x, y \rangle$ and $c_{\text{lin}} = -\langle x, y \rangle$ have the same minimizers as well (rescaling by $2 > 0$). Consequently, the Monge minimizers for

$$c_{\text{quad}}(x, y) = \|x - y\|^2 \quad \text{and} \quad c_{\text{lin}}(x, y) = -\langle x, y \rangle$$

coincide.

Remark (value relation). For any admissible T ,

$$\int \|x - T(x)\|^2 d\alpha = \int \|x\|^2 d\alpha + \int \|y\|^2 d\beta - 2 \int \langle x, T(x) \rangle d\alpha,$$

so minimizing the quadratic cost is equivalent to maximizing the correlation term $\int \langle x, T(x) \rangle d\alpha$; the optimal *map* is the same in both problems, although the optimal *values* differ by the constants depending only on α, β .

Exercise 10 (Quantile coupling: explicit computation).

Let $\mu = \mathcal{U}[0, 1]$ and let ν be supported on $[0, 1]$ with density $2y$. (a) Compute the cumulative distribution functions F_μ and F_ν , and their quantile functions F_μ^{-1} and F_ν^{-1} . (b) Using monotone transport in 1D, write the optimal map $T = F_\nu^{-1} \circ F_\mu$. (c) Compute $W_2^2(\mu, \nu)$; using both $\int_0^1 |x - T(x)|^2 dx$ and the 1D quantile formula $W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 dt$.

Useful hints

- (a) Integrate $2y$ from 0 to y , then invert.
- (b) Compose the target quantile with the source cumulative.
- (c) Expand $(x - \sqrt{x})^2$ and integrate its three terms.

Detailed solution

(a) For $\mu = \mathcal{U}[0, 1]$, $F_\mu(x) = 0$ for $x < 0$, $F_\mu(x) = x$ for $x \in [0, 1]$, $F_\mu(x) = 1$ for $x > 1$, and $F_\mu^{-1}(t) = t$ for $t \in [0, 1]$. For ν with density $2y$ on $[0, 1]$, $F_\nu(y) = 0$ for $y < 0$, $F_\nu(y) = y^2$ for $y \in [0, 1]$, $F_\nu(y) = 1$ for $y > 1$, and $F_\nu^{-1}(t) = \sqrt{t}$ for $t \in [0, 1]$. (b) The optimal map is $T = F_\nu^{-1} \circ F_\mu$, hence $T(x) = \sqrt{x}$ for $x \in [0, 1]$. (c) Using $\int_0^1 |x - T(x)|^2 dx$, compute $I = \int_0^1 (x - \sqrt{x})^2 dx = \int_0^1 (x^2 - 2x^{3/2} + x) dx = (1/3) - 2 \cdot (2/5) + (1/2) = 1/30$. Therefore $W_2^2(\mu, \nu) = 1/30$. Equivalently, by the quantile formula, $W_2^2(\mu, \nu) = \int_0^1 |t - \sqrt{t}|^2 dt = 1/30$.

Exercise 11 (Cumulative functions).

Let α be a probability measure on $\mathbb{R}$ and let its cumulative distribution function be

$$C_\alpha(x) = \alpha((-\infty, x]), \quad x \in \mathbb{R}.$$

Consider the pushforward by C_α , namely $(C_\alpha)_\# \alpha$.

- (1) Assume α is atomless (equivalently, C_α is continuous). This includes, but is not restricted to, measures with a Lebesgue density. Prove that

$$(C_\alpha)_\# \alpha = \mathcal{U}[0, 1].$$

In your proof, clearly point out *where* the continuity assumption is used.

- (2) What happens if α is purely discrete, say $\alpha = \sum_i p_i \delta_{x_i}$ with $\sum_i p_i = 1$ and $x_1 < x_2 < \dots$? Describe the law of $C_\alpha(X)$ for $X \sim \alpha$, and show that for all $t \in [0, 1]$,

$$\mathbb{P}(C_\alpha(X) \leq t) \leq t,$$

with equality at the jump values of C_α .

Useful hints

- (1) If U is uniform on $(0, 1)$, then $C_\alpha^{-1}(U)$ has law α . Use continuity to evaluate $C_\alpha(C_\alpha^{-1}(u))$.
- (2) At x_i the cumulative is the partial sum $s_i = \sum_{j \leq i} p_j$. Sum exactly those masses with $s_i \leq t$.

Detailed solution

(1) Write $F = C_\alpha$ and $Q(u) = \inf\{x : F(x) \geq u\}$ for $0 < u < 1$. For $U \sim \mathcal{U}(0, 1)$, the equivalence $Q(U) \leq x \iff U \leq F(x)$ proves $Q(U) \sim \alpha$. Continuity of F gives $F(Q(u)) = u$ for $0 < u < 1$, even when F has flat intervals. Consequently $F(Q(U)) = U$ almost surely, so $F_\# \alpha = \mathcal{U}[0, 1]$.

Continuity excludes jumps, not flat intervals. It is equivalent to absence of atoms, not to existence of a density; a singular atomless law also satisfies the conclusion. Endpoints $u = 0, 1$ are null events and need no finite quantile convention.

(2) At the ordered atoms, $F(x_i) = s_i$, so

$$F_\# \alpha = \sum_i p_i \delta_{s_i}, \quad \mathbb{P}(F(X) \leq t) = \sum_{i: s_i \leq t} p_i \leq t.$$

For a finite initial segment the sum equals its last partial sum; for an infinite one take a limit. Equality holds at $t = s_i$ (and at $0, 1$). This is often called a *super-uniform* random variable: its cumulative lies below the uniform cumulative.

Exercise 12 (Pushforward of a uniform law by a nonlinear map).

Let $\alpha = \mathcal{U}[-1, 1]$ and consider the map $T : \mathbb{R} \rightarrow \mathbb{R}$, $T(x) = x^2$.

- (1) Compute the law $\beta = T_\# \alpha$, i.e. find its density with respect to Lebesgue measure.
- (2) Check that this density integrates to 1 over its support.
- (3) Sketch the graph of the density and comment on its behaviour near 0 and 1.

Useful hints

- (1) Compute $\mathbb{P}(X^2 \leq y)$ before differentiating, or add the contributions from $\pm\sqrt{y}$.
- (2) Use the antiderivative $\sqrt{y}$.
- (3) Distinguish an integrable density singularity from an atom.

Detailed solution

(1) Let $X \sim \mathcal{U}[-1, 1]$ with density $f_X(x) = \frac{1}{2}$ on $[-1, 1]$. Define $Y = T(X) = X^2$. Then Y takes values in $[0, 1]$. For $y \in (0, 1)$, the preimage under T is $\{\pm\sqrt{y}\}$. By the standard change-of-variable formula for densities,

$$f_Y(y) = \sum_{x: x^2=y} \frac{f_X(x)}{|T'(x)|}.$$

Here $T'(x) = 2x$, so $|T'(\pm\sqrt{y})| = 2\sqrt{y}$. Each preimage contributes

$$\frac{f_X(\pm\sqrt{y})}{|T'(\pm\sqrt{y})|} = \frac{1/2}{2\sqrt{y}} = \frac{1}{4\sqrt{y}}.$$

Summing over $\pm\sqrt{y}$,

$$f_Y(y) = \frac{1}{4\sqrt{y}} + \frac{1}{4\sqrt{y}} = \frac{1}{2\sqrt{y}}, \quad y \in (0, 1].$$

Thus

$$f_Y(y) = \frac{1}{2\sqrt{y}} \text{ for } y \in (0, 1], \quad f_Y(y) = 0 \text{ otherwise.}$$

(2) Check normalization:

$$\int_0^1 \frac{1}{2\sqrt{y}} dy = [\sqrt{y}]_0^1 = 1.$$

So f_Y is a valid density.

(3) The graph of f_Y on $[0, 1]$ diverges like $1/(2\sqrt{y})$ as $y \downarrow 0$, showing a heavy concentration of mass near 0 (because many x in $[-1, 1]$ map to small squares). The density decreases monotonically to $1/2$ at $y = 1$.

Further exercises

Exercise 13 (Radial symmetry).

Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ be invariant under all orthogonal maps, and assume α is absolutely continuous. Define the radial cumulative function $F_\mu(r) = \mu(\{x : \|x\| \leq r\})$, $r \geq 0$, and its generalized inverse for $0 < u < 1$.

- (i) Show that the unique quadratic optimal map commutes with orthogonal maps: $Q \circ T \circ Q^{-1} = T$ α -a.e. Show that it has a radial representative $T(x) = t(\|x\|)x/\|x\|$ for α -a.e. $x \neq 0$, with t nonnegative and nondecreasing on the relevant radii.
- (ii) Prove $t(r) = F_\beta^{-1}(F_\alpha(r))$ for almost every source radius. Does $F_\beta(t(r)) = F_\alpha(r)$ still hold if β has an atom on a sphere?
- (iii) Specialize to uniform laws on balls of positive radii R_α, R_β . Give the map and its squared cost.

Useful hints

- (i) Conjugate an optimal map by Q and use uniqueness. To avoid choosing exceptional sets for all Q simultaneously, construct a radial optimizer directly using $\|x - y\| \geq |||x| - |y|||$.
- (ii) Apply one-dimensional increasing rearrangement to the laws of the radii. The target radial cumulative may have jumps.
- (iii) The source radial cumulative is $(r/R_\alpha)^d$. Compute $\mathbb{E}\|X\|^2$ in polar coordinates.

Detailed solution

(i) Orthogonal invariance of the marginals and cost makes $Q \circ T \circ Q^{-1}$ another optimizer. Brenier uniqueness gives the stated equality for each fixed Q .

For a direct construction, let $R = \|X\|$ for $X \sim \alpha$. Its law is atomless. By radial invariance, $X = RU$ where U is uniform on the unit sphere and independent of R (in dimension one the sphere consists of $\{-1, 1\}$). Put $t = F_\beta^{-1} \circ F_\alpha$. Then $t(R)$ has the target radial law, and $t(R)U$ has law β , including any target mass at the origin.

Every coupling of α, β induces a coupling of their radii and satisfies

$$\int \|x - y\|^2 d\pi \geq \int |||x| - |y|||^2 d\pi \geq \int |r - t(r)|^2 d(\|\cdot\|_\# \alpha)(r).$$

The constructed map attains both inequalities. It is therefore the unique optimal map, with the claimed radial form. Its radial profile is nonnegative and nondecreasing. The value at $x = 0$ may be set arbitrarily since $\alpha(\{0\}) = 0$.

(ii) The preceding construction proves the quantile formula. More precisely, for $u = F_\alpha(r) \in (0, 1)$,

$$F_\beta(t(r)^-) \leq u \leq F_\beta(t(r)).$$

Equality on the right requires continuity at the target radius and is false in general: if β is uniform on the unit sphere, then $t(r) = 1$ for almost every source radius, while $F_\beta(1) = 1$. Mass conservation means $t_\#(\|\cdot\|_\# \alpha) = \|\cdot\|_\# \beta$, not a pointwise equality of cumulative functions at jumps.

(iii) For uniform balls, matching $(r/R_\alpha)^d$ with $(t/R_\beta)^d$ gives $T(x) = (R_\beta/R_\alpha)x$. Since $\mathbb{E}\|X\|^2 = dR_\alpha^2/(d+2)$,

$$\mathcal{W}_2^2(\alpha, \beta) = \frac{d}{d+2}(R_\alpha - R_\beta)^2.$$

This also follows directly because a positive dilation is the gradient of a convex quadratic.